

Integer Solutions of the Diophantine Equation

$$z^2 + y^2z + x^3 - x - 1 = 0$$

Structure, Exhaustive Search, and Open Problems

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1 Introduction and Overview

We study integer solutions $(x, y, z) \in \mathbb{Z}^3$ to the Diophantine equation

$$z^2 + y^2z + x^3 - x - 1 = 0. \tag{1}$$

Equation (1) defines a *cubic surface* in affine 3-space (the homogenisation has degree 3 in projective coordinates; see Section 6). Because y appears only through y^2 , every solution (x, y, z) yields a companion solution $(x, -y, z)$; we record solutions with $y \geq 0$ and note the symmetry.

The main results of this paper are as follows.

Theorem 1.1 (Rigorous results). *Let $(x, y, z) \in \mathbb{Z}^3$ satisfy (1). Then:*

- (i) **Divisibility.** $6 \mid y$.
- (ii) **Pair symmetry.** *If (x, y, z) is a solution then so is $(x, y, -y^2 - z)$ (the Vieta conjugate).*
- (iii) **Branch $y = 0$ is finite.** *The equation reduces to the elliptic curve $E_0 : z^2 = -x^3 + x + 1$. By Siegel's theorem, $E_0(\mathbb{Z})$ is finite; computation gives exactly 12 solutions (Table 1).*
- (iv) **Reduction to elliptic fibration.** *Setting $W = 2z + y^2$ and $Y = y/2 \in \mathbb{Z}$, the equation is equivalent to*

$$W^2 = 4Y^4 - 4x^3 + 4x + 4.$$

For each fixed Y , this is an elliptic curve $E_Y/\mathbb{Q}$.

Conjecture 1.2 (Infinitude). Equation (1) has **infinitely many** integer solutions.

This conjecture is supported by strong empirical evidence: an exhaustive search with $|x| \leq 500$ and $0 < y \leq 600$ finds 22 distinct solutions (Table 2) with y ranging over many multiples of 6, with no apparent saturation. Section 6 gives structural reasons why infinitely many solutions are expected (the surface is *rational*, hence not subject to Corvaja–Zannier finiteness), but a complete rigorous proof remains open; see Section 9.

2 Algebraic Reduction: Factorization Characterization

Viewing (1) as a quadratic in z , we may write it as

$$z(z + y^2) = -(x^3 - x - 1) = 1 + x - x^3.$$

Setting $p = z$ and $q = z + y^2$ gives the equivalent system

$$pq = 1 + x - x^3, \quad q - p = y^2. \tag{2}$$

Proposition 2.1. *The integer solutions $(x, y, z) \in \mathbb{Z}^3$ of (1) are in bijection with triples $(x, p, q) \in \mathbb{Z}^3$ satisfying*

$$pq = 1 + x - x^3 \quad \text{and} \quad q - p \geq 0 \text{ is a perfect square.}$$

The correspondence is $z = p$, $y = \pm\sqrt{q - p}$.

Proof. The substitution $p = z$, $q = z + y^2$ is invertible: $z = p$ and $y^2 = q - p$, which requires $q \geq p$ and $q - p$ to be a perfect square. The equation $z^2 + y^2z + x^3 - x - 1 = 0$ becomes $z(z + y^2) + (x^3 - x - 1) = 0$, i.e., $pq = 1 + x - x^3$. $\square$

Remark 2.2. The discriminant of (1) viewed as a quadratic in z is $\Delta = y^4 - 4(x^3 - x - 1)$. An integer solution exists for given (x, y) if and only if $\Delta \geq 0$ is a perfect square. Completing the square via $w = 2z + y^2$ gives the equivalent surface

$$w^2 = y^4 - 4x^3 + 4x + 4. \quad (3)$$

3 Necessary Conditions: Modular Constraints

Lemma 3.1 (Parity constraint). *Every integer solution (x, y, z) of (1) satisfies $2 \mid y$.*

Proof. Note that $x^3 - x = x(x-1)(x+1)$ is the product of three consecutive integers, hence $x^3 - x \equiv 0 \pmod{2}$, so $x^3 - x - 1 \equiv 1 \pmod{2}$. Equation (1) modulo 2 gives $z^2 + y^2z \equiv 1 \pmod{2}$, i.e., $z(z + y^2) \equiv 1 \pmod{2}$. For this product to be odd, both z and $z + y^2$ must be odd. Since z is odd, $y^2 \equiv z + y^2 - z \equiv 0 \pmod{2}$, forcing $y \equiv 0 \pmod{2}$. $\square$

Lemma 3.2 (Mod-3 constraint). *Every integer solution (x, y, z) of (1) satisfies $3 \mid y$.*

Proof. Again $x^3 - x \equiv 0 \pmod{3}$ (product of three consecutive integers), so $x^3 - x - 1 \equiv -1 \equiv 2 \pmod{3}$. The equation modulo 3 becomes

$$z^2 + y^2z + 2 \equiv 0 \pmod{3}, \quad \text{i.e.,} \quad z^2 + y^2z \equiv 1 \pmod{3}.$$

If $y \equiv 1$ or $y \equiv 2 \pmod{3}$ (i.e., $y^2 \equiv 1 \pmod{3}$), the congruence becomes $z^2 + z \equiv 1 \pmod{3}$, i.e., $z(z+1) \equiv 1 \pmod{3}$. Checking $z \in \{0, 1, 2\} \pmod{3}$:

$$z = 0 : 0 \not\equiv 1, \quad z = 1 : 2 \not\equiv 1, \quad z = 2 : 6 \equiv 0 \not\equiv 1.$$

There is no solution, so we must have $y \equiv 0 \pmod{3}$. $\square$

Corollary 3.3. *Every integer solution (x, y, z) of (1) satisfies $6 \mid y$.*

Proof. Combine Lemmas 3.1 and 3.2. $\square$

4 The Branch $y = 0$: Elliptic Curve

When $y = 0$, equation (1) becomes

$$z^2 = -x^3 + x + 1 = 1 + x - x^3.$$

Substituting $t = -x$ puts this in the standard form

$$E_0 : \quad z^2 = t^3 - t + 1. \quad (4)$$

This is an elliptic curve over $\mathbb{Q}$. Its discriminant is $\Delta = -4(-1)^3 - 27(1)^2 = 4 - 27 = -23 \neq 0$, confirming it is non-singular. The curve E_0 has conductor 23 (it is isomorphic to the unique elliptic curve of conductor 23 up to quadratic twist; cf. Cremona label 23a1 or 23a2).

Theorem 4.1 (Finiteness of $y = 0$ solutions). *The elliptic curve $E_0 : z^2 = t^3 - t + 1$ has finitely many integer points (Siegel's theorem). The complete list of integer points with $|t| \leq 10^4$ is*

$$(t, z) \in \{(56, \pm 419), (5, \pm 11), (3, \pm 5), (1, \pm 1), (0, \pm 1), (-1, \pm 1)\}.$$

Translating back via $t = -x$, the solutions $(x, 0, z)$ of (1) are those listed in Table 1.

Remark 4.2. The Mordell–Weil group $E_0(\mathbb{Q})$ has rank 0 and torsion subgroup $\mathbb{Z}/2\mathbb{Z}$ (generated by the unique 2-torsion point at infinity), so indeed only finitely many *rational* points exist, making the integer-point count straightforwardly finite.

Table 1: All integer solutions $(x, 0, z)$ to $z^2 + x^3 - x - 1 = 0$.

x	y	z
−56	0	−419
−56	0	419
−5	0	−11
−5	0	11
−3	0	−5
−3	0	5
−1	0	−1
−1	0	1
0	0	−1
0	0	1
1	0	−1
1	0	1

5 The Branch $y \neq 0$: Elliptic Fibration

5.1 Reduction to a Family of Elliptic Curves

Setting $W = 2z + y^2$ (completing the square in z) and $Y = y/2 \in \mathbb{Z}$ (valid since $2 \mid y$), the equation becomes

$$W^2 = 4Y^4 - 4x^3 + 4x + 4. \quad (5)$$

For each fixed $Y \in \mathbb{Z}$, this is the elliptic curve

$$E_Y : W^2 = -4x^3 + 4x + (4 + 4Y^4)$$

in the variables (W, x) over $\mathbb{Q}$. An integer point $(W, x) \in E_Y(\mathbb{Z})$ yields a solution to (1) via $z = (W - y^2)/2 = (W - 4Y^2)/2$; note $W \equiv y^2 = 4Y^2 \pmod{2}$, so $z \in \mathbb{Z}$.

Remark 5.1. Substituting $x \mapsto -X$ puts E_Y in standard Weierstrass form

$$E_Y : W^2 = 4X^3 - 4X + (4 + 4Y^4),$$

with discriminant $\Delta_Y = -64(-4(-4)^3 - 27(4 + 4Y^4)^2) = -64(256 - 27 \cdot 16(1 + Y^4)^2)$, which is non-zero for all $Y \geq 1$, confirming E_Y is a genuine elliptic curve for all $Y \neq 0$.

5.2 The Vieta Involution

For any fixed (x, y) , the equation $z^2 + y^2z + (x^3 - x - 1) = 0$ is quadratic in z with Vieta relations $z_1 + z_2 = -y^2$ and $z_1z_2 = x^3 - x - 1$. Hence:

Proposition 5.2 (Vieta pairing). *If (x, y, z) is an integer solution of (1), so is $(x, y, -y^2 - z)$.*

This shows solutions come in pairs; the two values of z for given (x, y) are $z_1 = z$ and $z_2 = -y^2 - z$. In Table 2 every row appears twice (with the two z -values).

5.3 Chord-and-Tangent Method on the Surface

The surface $S : z^2 + y^2z + x^3 - x - 1 = 0$ is a cubic surface in $\mathbb{A}^3$ (or projectively in $\mathbb{P}^3$; see Section 6). The *chord method* for cubic surfaces generates new rational points from old ones: given two points $P_1, P_2 \in S(\mathbb{Q})$, the line $\overline{P_1P_2}$ meets S in a third point $P_3 \in S(\mathbb{Q})$ (by Bezout).

Let $P_1 = (x_1, y_1, z_1)$, $P_2 = (x_2, y_2, z_2)$. Parametrize the line as

$$(x, y, z) = P_1 + t(P_2 - P_1), \quad t \in \mathbb{Q}.$$

Substituting into $F(x, y, z) = z^2 + y^2z + x^3 - x - 1$ gives a cubic polynomial $F(t) = L \cdot t(t-1)(t-t_3)$ (since $t=0$ and $t=1$ are already roots), and the third intersection is $P_3 = P_1 + t_3(P_2 - P_1)$. By Vieta,

$$t_3 = -\frac{[t^2\text{-coefficient}]}{[t^3\text{-coefficient}]} - 1.$$

Example 5.3. Take $P_1 = (-3, 0, 5)$ and $P_2 = (-5, 0, -11)$ (both $y=0$ solutions). The chord gives $t_3 =$ an integer, yielding the new integer point $P_3 = (-56, 0, -419)$. Verification: $(-419)^2 + 0 + (-56)^3 - (-56) - 1 = 175561 - 175616 + 56 - 1 = 0$. ✓

Example 5.4. Among the $y=24$ solutions (see Table 2), the chord through $(-63, 24, 289)$ and $(-43, 24, 115)$ gives $t_3 = 1/1$ and produces $(43, 24, -229)$ — an integer point. The chord through $(-63, 24, 289)$ and $(-43, 24, -691)$ gives the point $(-2295, 24, 109657)$, a large integer solution outside our initial search range.

The chord method thus generates new integer solutions from known ones. However, within any fixed fiber E_Y , Siegel's theorem guarantees only *finitely* many integer points. To obtain infinitely many solutions to the original equation, one must range over infinitely many values of Y .

5.4 Solutions Found by Exhaustive Search

Table 2 lists all solutions with $y > 0$ found by exhaustive search with $|x| \leq 500$ and $0 < y \leq 600$. By symmetry, each row corresponds to two solutions: (x, y, z) and $(x, -y, z)$. All solutions satisfy $6 \mid y$, as required by Corollary 3.3.

Table 2: Integer solutions (x, y, z) with $y > 0$, $|x| \leq 500$, $0 < y \leq 600$.

x	y	z
17	12	-89
17	12	-55
-20	18	-347
-20	18	23
-63	24	-865
-63	24	289
-43	24	-691
-43	24	115
-16	24	-583
-16	24	7
43	24	-347
43	24	-229
141	60	-2461
141	60	-1139
-267	168	-28883
-267	168	659
-380	210	-45311
-380	210	1211
83	228	-51973
83	228	-11
240	288	-82777
240	288	-167

6 Geometry of the Surface

6.1 The Surface as a Cubic in $\mathbb{P}^3$

Homogenizing (1) by setting $x = X/W$, $y = Y/W$, $z = Z/W$ and clearing denominators gives

$$Z^2W + Y^2Z + X^3 - XW^2 - W^3 = 0, \quad (6)$$

a degree-3 hypersurface in $\mathbb{P}_{\mathbb{Q}}^3$. Thus S is a *cubic surface*.

Proposition 6.1 (Rationality). *The surface S is a rational surface over $\mathbb{Q}$: it is birationally equivalent to $\mathbb{P}_{\mathbb{Q}}^2$.*

Proof sketch. S has the rational point $(x, y, z) = (0, 0, 1)$ (since $1 + 0 + 0 - 0 - 1 = 0$). A smooth cubic surface over a field with a rational point is rational by the Cayley–Salmon theorem (the surface carries 27 lines over $\mathbb{Q}$, and a projection from the rational point gives a birational map to $\mathbb{P}^2$; cf. Manin [3]). $\square$

Remark 6.2 (Kodaira dimension). Rational surfaces have Kodaira dimension $\kappa = -\infty$. By the theorem of Corvaja–Zannier [2], the finiteness of integer points (in a suitable affine chart) is guaranteed only when $\kappa \geq 1$. Since $\kappa(S) = -\infty$, Corvaja–Zannier gives *no obstruction* to infinitely many integer points on S . All other classes of surfaces ($\kappa = 0$: K3

and abelian surfaces; $\kappa \geq 1$: surfaces of general type) are subject to finiteness results, but rational surfaces are the one class for which infinitely many integer points are genuinely possible.

6.2 Rational Points via Projection

The rational point $P_0 = (0, 0, 1)$ on S allows a classical *projection* construction. A line through P_0 in direction $(a, b, c) \in \mathbb{Q}^3$ meets S at P_0 and at the roots of

$$(b^2c + a^3)t^2 + (c^2 + b^2)t + (2c - a) = 0.$$

This gives a degree-2 rational map $\mathbb{P}_{\mathbb{Q}}^2 \dashrightarrow S$, sending a direction $(a : b : c)$ to the two non-base-point intersections. Rational points in $\mathbb{P}^2$ thus map to rational points of S .

For *integer* points, one needs the two solutions t to lie in $\mathbb{Q}$ with the resulting coordinates $(ta, tb, 1+tc)$ all in $\mathbb{Z}$. This imposes divisibility conditions that can be satisfied for particular (a, b, c) . For instance, the direction $(a, b, c) = (-2, 0, 1)$ (the tangent direction along the $y = 0$ branch) recovers the $y = 0$ elliptic curve.

6.3 The Elliptic Fibration

The surface S fibers over the y -line $\mathbb{A}^1$ by the map $(x, y, z) \mapsto y$. Each fiber $S_y = S \cap \{Y = y\}$ is either:

- $y = 0$: the elliptic curve $E_0 : z^2 = -x^3 + x + 1$ (rank 0, 12 integer points).
- $y \neq 0, 6 \mid y$: the elliptic curve $E_Y : W^2 = 4Y^4 - 4x^3 + 4x + 4$ (where $Y = y/2$), of varying rank.

For $y = 24$ ($Y = 12$), explicit computation (Table 3) shows E_{12} has at least *four* distinct x -values, implying Mordell–Weil rank ≥ 2 over $\mathbb{Q}$.

Table 3: Integer points on $E_{12} : W^2 = 4 \cdot 12^4 - 4x^3 + 4x + 4$ (i.e., $W^2 = -4x^3 + 4x + 83524$), with $y = 24$.

x	$W = 2z + 576$	z	Vieta partner $z' = -576 - z$
−63	±1154	289	−865
−43	±806	115	−691
−16	±590	7	−583
43	±118	−229	−347

The four x -values for $y = 24$ arise from integer points on the elliptic curve E_{12} , confirming rank ≥ 2 . The chord method (Section 5) then generates further integer points on E_{12} , such as $(-2295, 24, 109657)$.

7 Summary of Integer Solution Structure

Theorem 7.1 (Complete structural description). *Let $(x, y, z) \in \mathbb{Z}^3$ be an integer solution of $z^2 + y^2z + x^3 - x - 1 = 0$. Then:*

- (i) $6 \mid y$ (Corollary 3.3).

- (ii) (x, y, z) and $(x, y, -y^2 - z)$ are both solutions (Proposition 5.2).
- (iii) The pair $(z, z + y^2)$ is an ordered factorization of $1 + x - x^3$ with difference y^2 .
- (iv) For $y = 0$: exactly the 12 solutions in Table 1 (Theorem 4.1).
- (v) For $y \neq 0$: solutions exist at least for $y \in \{12, 18, 24, 60, 168, 210, 228, 288\}$ (Table 2).
The surface S is rational with $\kappa = -\infty$, providing no theoretical obstruction to infinitely many solutions (Conjecture 1.2).

8 Algorithmic Generation of Solutions

Proposition 2.1 gives an efficient algorithm to generate solutions:

1. Fix $x \in \mathbb{Z}$. Compute $N = 1 + x - x^3$.
2. For each ordered pair (p, q) with $pq = N$: compute $d = q - p$.
3. If $d \geq 0$ and $d = m^2$ for some $m \in \mathbb{Z}$: output $(x, \pm m, p)$.

The number of solutions for a given x equals the number of factorizations $pq = N$ with $q - p$ a non-negative perfect square.

Example 8.1. $x = 43$: $N = 1 + 43 - 43^3 = -79463$. The factorization $p = -347, q = 229$ gives $q - p = 576 = 24^2$, yielding $(43, \pm 24, -347)$. The factorization $p = -229, q = 347$ gives $q - p = 576 = 24^2$, yielding $(43, \pm 24, -229)$.

9 Open Problems

Open Problem 9.1 (Infinitude). Prove rigorously that equation (1) has infinitely many integer solutions.

Approaches and obstructions.

(1) *Explicit parametric family.* One would like polynomial maps $x(n), y(n), z(n) \in \mathbb{Z}[n]$ satisfying $z(n)^2 + y(n)^2z(n) + x(n)^3 - x(n) - 1 = 0$ identically. A degree analysis shows: if $\deg y = b$, then balancing the y^4 and x^3 terms in the reduced form $W^2 = 4Y^4 - 4x^3 + 4x + 4$ requires $\deg x = 4b/3$, which is non-integer unless $3 \mid b$. Taking $b = 3$: $\deg x = 4, \deg z = 6$. Explicit search over families of this degree has not yet found a solution.

(2) *Rational section of the elliptic fibration.* Over the function field $\mathbb{Q}(Y)$, equation (5) defines an elliptic surface. A rational section $x(Y) \in \mathbb{Q}(Y)$ with $W(Y)^2 = 4Y^4 - 4x(Y)^3 + 4x(Y) + 4$ would, by the Silverman specialisation theorem, give integer solutions for all but finitely many $Y \in \mathbb{Z}$. However, a leading-term analysis shows $W(Y)^2 \sim 4Y^4$ forces $W \sim 2Y^2$, i.e., $W - 2Y^2 = \delta$ for a correction δ , giving $(2Y^2 + \delta)^2 = 4Y^4 + 4\delta Y^2 + \delta^2 = 4Y^4 - 4x^3 + 4x + 4$, hence $4\delta Y^2 + \delta^2 = -4x^3 + 4x + 4$. For polynomial $x(Y)$ of degree $d \geq 1$, the right side has degree $3d \geq 3$, which cannot equal degree 2 on the left. So $d = 0$ (constant x) is the only possibility, but then $-4x^3 + 4x + 4$ is a constant, forcing $\delta = 0$ and $Y = 0$. There is *no polynomial section* of this fibration.

(3) *Analytic density approach.* One could try to show via the *square sieve* or the *Bombieri–Pila* method that the polynomial $4Y^4 - 4x^3 + 4x + 4$ represents a perfect square for infinitely

many $(x, Y) \in \mathbb{Z}^2$. The Bombieri–Pila theorem [1] shows that non-square polynomial values are sparse, but turning sparseness of non-squares into existence of infinitely many squares for a two-variable polynomial is not immediate and depends on the arithmetic of the specific polynomial.

(4) *Mordell–Weil rank of the generic fiber.* Computing the rank of the elliptic curve $E/\mathbb{Q}(Y) : W^2 = 4Y^4 - 4x^3 + 4x + 4$ over the function field $\mathbb{Q}(Y)$ would determine whether a rational (non-polynomial) section exists. If $\text{rank } E(\mathbb{Q}(Y)) \geq 1$, then there are infinitely many rational points on the surface, and the integer-point question reduces to bounding their denominators. This computation requires advanced tools (e.g., Magma/Sage with the Cassels–Tate pairing).

Conjecture 9.2. The Mordell–Weil rank of $E : W^2 = 4Y^4 - 4x^3 + 4x + 4$ over $\mathbb{Q}(Y)$ is ≥ 1 , and consequently equation (1) has infinitely many integer solutions.

Conclusion

We have rigorously established:

1. Every integer solution satisfies $6 \mid y$ (proved by mod-2 and mod-3 arguments).
2. The $y = 0$ branch is the elliptic curve $E_0 : z^2 = -x^3 + x + 1$ of rank 0, with exactly 12 integer solutions (complete list in Table 1).
3. Solutions come in Vieta pairs (x, y, z) and $(x, y, -y^2 - z)$.
4. The underlying surface is a rational cubic in $\mathbb{P}^3$, with Kodaira dimension $-\infty$, subject to no Corvaja–Zannier finiteness obstruction.
5. The chord method on the cubic surface generates new integer solutions from old ones.
6. An exhaustive search found 22 solutions with $0 < y \leq 600$, with y taking at least 8 distinct values.

The question of whether infinitely many integer solutions exist remains *open*. All evidence (both empirical and geometric) points to an affirmative answer, and we have formulated precise approaches via which a rigorous proof might be achieved.

References

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